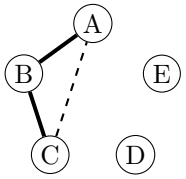
Worksheet: Hamiltonian Circuits & Sorted Edges / Cheapest Link

Group Names: _____

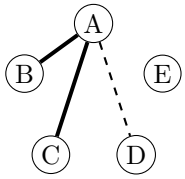
Sorted Edges / Cheapest Link: Sort the edges from cheapest to most expensive. Add the next cheapest edge to your circuit *unless* (1) it closes the circuit too soon, or (2) it creates a degree 3 vertex. Break ties by choosing the alphabetically smallest edge.

1. In each picture, the thick edges have already been chosen by the Sorted Edges algorithm, and the dashed edge is the next cheapest edge. Should the dashed edge be added? If not, which rule does it break?

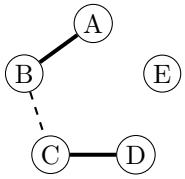
(a)



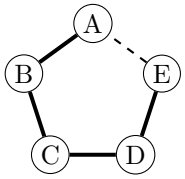
(b)



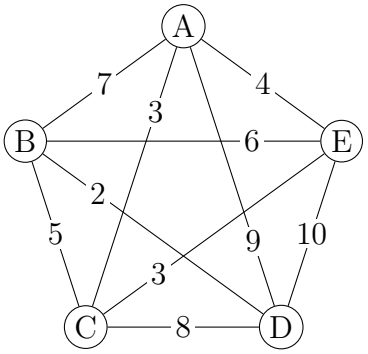
(c)

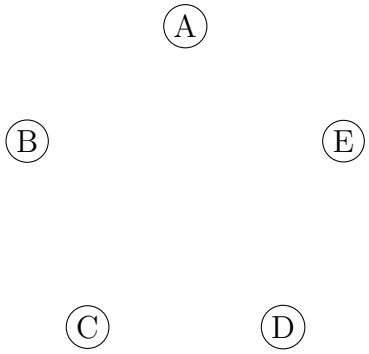


(d)



2. Apply the Sorted Edges algorithm to the graph below to find a Hamiltonian circuit. Draw in the edges, labeled with their weight, as you add them on the empty graph, and keep track of each step in the table. Break ties by choosing the alphabetically smaller edge.





Sorted edges	weight	used? (or why not)
BD	2	
AC	3	
CE	3	
AE	4	
BC	5	
BE	6	
AB	7	
CD	8	
AD	9	
DE	10	

List the vertices of the Hamiltonian circuit, starting at vertex A.

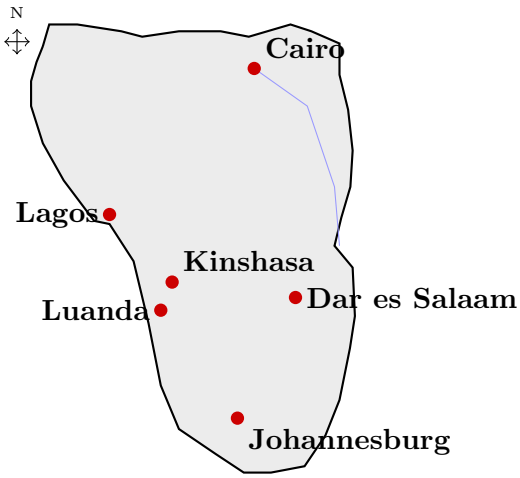
_____

Total weight of the circuit:

_____

3. On the previous worksheet, you used Nearest Neighbor to plan a tour of the six most populous cities in Africa. Here is the same data, with the edges already sorted by distance (in km).

Sorted edges	distance	used?/why not
Kinshasa – Luanda	554	
Lagos – Kinshasa	1,798	
Lagos – Luanda	2,027	
Dar es Salaam – Johannesburg	2,461	
Luanda – Johannesburg	2,484	
Kinshasa – Dar es Salaam	2,651	
Kinshasa – Johannesburg	2,781	
Luanda – Dar es Salaam	2,870	
Cairo – Lagos	3,913	
Cairo – Kinshasa	4,179	
Cairo – Dar es Salaam	4,184	
Lagos – Dar es Salaam	4,242	
Lagos – Johannesburg	4,508	
Cairo – Luanda	4,732	
Cairo – Johannesburg	6,264	



(a) Use the Sorted Edges algorithm to find a Hamiltonian circuit. List the cities of your circuit in order, starting and ending at Cairo.

(b) What is the total distance of your circuit? _____

(c) Nearest Neighbor starting at Johannesburg gave a tour of 17,870 km. How does your Sorted Edges tour compare?

4. The Sorted Edges algorithm can't always find a Hamiltonian circuit at all. Construct a small example of a graph that *has* a Hamiltonian circuit, but Sorted Edges doesn't find it.